

ORMatch: local reviewer matching for OR/MS papers

Technical note

September 2026

1 What the tool does

ORMatch reads a manuscript PDF. It returns a ranked list of candidate reviewers. Each candidate comes with the papers that support the match. Candidates with a possible conflict of interest are flagged, and the reason is shown.

The tool has two parts. The *index* is public. It holds titles, abstracts, authorships, reference lists and precomputed embeddings of published papers. It is built offline and shipped as a versioned archive. The *query* is private. Text extraction, embedding, search and ranking run on the user's machine. The manuscript, its embedding and its metadata never leave the machine. The command `ormatch verify-offline` runs the full pipeline with network sockets disabled and fails if any network call is attempted. The tool uses the network only to download the index and the model weights once.

2 Data

2.1 Source

All bibliographic data come from OpenAlex (<https://openalex.org>, CC0 licence). The tool pulls works through the OpenAlex REST API with cursor pagination. Cursor pagination has no result cap. The filter is: publication year 2014 or later, and primary location in one of 29 venues.

The venues are: Operations Research, Management Science, M&SOM, Mathematics of Operations Research, Transportation Science, INFORMS Journal on Computing, Stochastic Systems, Mathematical Programming, SIAM Journal on Optimization, SIAM Journal on Control and Optimization, Production and Operations Management, Journal of Operations Management, Decision Sciences, Naval Research Logistics, IIE Transactions, IIE Transactions, European Journal of Operational Research, Operations Research Letters, Computers & Operations Research, Annals of Operations Research, Omega, International Journal of Production Economics, International Journal of Production Research, Computers & Industrial Engineering, Transportation Research Part B, Transportation Research Part E, Journal of Optimization Theory and Applications, Journal of the Operational Research Society, and Networks.

Some works list a venue only as a non-primary location. The tool includes these works only if they have a DOI. This rule removes RePEc and repository records that OpenAlex attributes to the journals by mistake.

Each pull is resumable. The tool stores the cursor, the counts and the filter next to the raw JSONL file. The script `check.pull.py` compares the stored counts with the live OpenAlex counts and reports any difference.

2.2 Fields

For each work the tool keeps: title, abstract, year, venue, DOI, open-access URL, citation count, the full author list with OpenAlex author IDs and first-listed institution, and the list of referenced works. OpenAlex stores abstracts as an inverted index. The tool rebuilds the plain text from it.

2.3 Missing abstracts

OpenAlex has no abstracts for most Elsevier papers. Before backfill, EJOR had abstracts for 17% of its papers and ORL for 11%. The tool requests missing abstracts from the Semantic Scholar Graph API in batches of 500 DOIs. Crossref returns almost no abstracts for Elsevier papers, so the Crossref pass is optional. A pass against the Elsevier Article Retrieval API is implemented. It needs an institutional API key and has not been run.

2.4 Side tables

- **Author statistics.** Works count, citation count and h -index for each author, from the OpenAlex authors endpoint.
- **Editorial boards.** The publishers’ board pages block scripted access. The tool reads yearly Internet Archive snapshots (2016–2026) of 14 journals’ board pages instead. It parses role headings and names. Each row records journal, role, person, first year seen and last year seen.
- **Academic genealogy.** Advisor and student links from the Mathematics Genealogy Project, fetched by name on request.
- **Department rosters.** Faculty lists from 167 OR, IE and OM departments in 25 countries, scraped where the sites permit it.

papers, 29 venues, 2014–2026	74,127
with a title (embedded)	73,643
with an abstract	46,302 (62.5%)
with an open-access PDF link	23,554 (31.8%)
distinct authors	101,183
editorial-board rows / people	2,667 / 1,964
with a role seen in 2025 or 2026	1,323
add-on collection stats-ml	8,621 papers

Table 1: Index contents on 2026-09-20. The 27,703 missing abstracts are almost all Elsevier papers.

2.5 Add-on collections

Only 29% of the references in core papers point to core venues. The tool can build additional collections as separate index directories. Each directory holds its own embeddings and tables. Defined collections are statistics and machine learning, economics and finance, and algorithms. The user can search the core and any add-ons together. An author who appears only in an add-on is proposed only if they have at least one link to the core literature. A link is a core paper, a citation to a core paper, or a citation from a core paper.

3 Embeddings

The encoder is SPECTER2: `allenai/specter2.base` with the `proximity` adapter. It is a SciBERT model trained with a citation-based contrastive objective on titles and abstracts. The input for each paper is the title, the separator token, and the abstract. If the abstract is missing, the input is the title only. The input is truncated to 512 tokens. The embedding is the final-layer [CLS] vector, normalised to unit length. Embeddings have 768 dimensions and are stored in float16. The full corpus embeds in about 200 seconds on one RTX 4090.

SciNCL (`malteos/scincl`) is an alternative encoder. TF-IDF with SVD is a fallback that needs no deep-learning libraries. The index records which encoder built it. A query always uses the same encoder as the index.

The manuscript follows the same path. `pypdfium2` extracts the text of the first pages. A heuristic finds the title and the abstract. The title is the leading lines before the word “Abstract” that do not look like an author line. The abstract runs from “Abstract” to “Introduction” or “Keywords”. The same encoder maps title and abstract to the query vector q .

The tool also parses the bibliography from the last pages. It splits entries where a line starts like an author list and the previous text ends a sentence. It reads DOIs with a regular expression. If there is no DOI, it takes the sentence after the year as the title. It matches entries to indexed papers by DOI, then by normalised title.

4 Matching

Let s_i be the cosine similarity between the manuscript vector q and the embedding of paper i . Let P_a be the indexed papers of author a . The score of author a is a sum of terms. All terms are in cosine units.

$$\begin{aligned} \text{score}(a) = & (1 - \lambda) \max_{i \in P_a} s_i + \lambda \text{mean}_{\text{top-}k}(s_i) + w_{\text{cite}}(1 - 0.5^{n_{\text{cited}}(a)}) + w_{\text{seed}}(\sigma_a - \text{med } \sigma) + w_{\text{early}} \phi(\text{works}_a) \\ & - w_{\text{vol}}(\mathbb{E}[\max \text{ of } |P_a| \text{ random } s] - \mu_s) - w_{\text{ed roles}_a} - w_{\text{sen}} [\log(1 + \text{works}_a) - \text{med}]_+. \end{aligned}$$

Text similarity. λ mixes the author’s best paper with the mean of their top- k papers. Max pooling and mean pooling are the two standard choices. OpenReview uses max by default. The ACL matcher uses the top three cosines with weights $1, \frac{1}{2}, \frac{1}{3}$.

Citation bonus. $n_{\text{cited}}(a)$ is the number of the author’s indexed papers that the manuscript cites. The bonus saturates. This term needs no full text on the index side.

Seeds. The user can name reviewers they already have in mind. σ_a is the cosine between author a ’s profile and the closest seed’s profile. A profile is the mean embedding of an author’s papers. Profile cosines cluster near 1, so the term uses the deviation from the candidate-pool median. Seeds are removed from the output.

Volume correction. An author with n papers has n chances at a high cosine. The term subtracts the expected best-of- n cosine of n random papers, computed from the query’s own similarity distribution with a normal approximation.

Editorial and seniority penalties. Current editorial roles count 1. Past roles count 0.5. The seniority penalty grows with log works count above the median author.

Early-career bonus. ϕ is 1 for one work and falls linearly in log works to 0 at a cutoff.

Recency (optional). s_i is shrunk toward the query mean μ_s by $0.5^{\text{age}/h}$. Multiplying the raw cosine does not work: dense-encoder cosines all lie in about $[0.7, 1]$, so the weight would dominate the topic.

The scoring loop is exact but runs on a candidate pool. A vectorised pass computes each author’s best paper. The pool is the top 2,000 authors (or $100n$ for a list of n), plus every author who is cited or seeded. Penalties only lower scores, so no author outside the pool can rank above the pool. Re-ranking 100,000 authors takes about 40 ms.

If the user sets a diversity value d , the tool re-orders the pool by maximal marginal relevance. It picks the candidate that maximises $(1 - d)\text{score}(a) - d\max_b \cos(a, b)$, where b ranges over the seeds and the candidates already picked. This spreads the list over different neighbourhoods.

Conflicts of interest. The user enters the manuscript’s authors. The tool resolves each name to index authors by normalised name, first exact, then surname plus initials. Ambiguous names are reported. The tool flags: co-authors on papers within a window (default 5 years), people at the same institution, genealogy links, and names on the user’s declared conflict list. The declared list persists between sessions. Flags show the reason. An exclude mode removes flagged candidates instead. Name resolution is fuzzy, so flagging is the default.

5 Evaluation

The protocol is leave-one-out. Sample 200 indexed papers. For each paper: mask it, use its embedding as the manuscript, use its OpenAlex reference list as the bibliography, rank all authors, and record the ranks of its true authors. Authors whose only paper is the held-out paper cannot be found and are excluded. This protocol rewards finding the actual authors among 100,000 candidates. It favours prolific authors. It is a proxy for fit, not a measure of fit.

index	configuration	MRR	recall@10	recall@20
17 venues, SciNCL	$\lambda=0.3, k=3$	0.189	0.210	0.273
17 venues, SPECTER2	$\lambda=0.3, k=3$	0.193	0.259	0.303
17 venues, SPECTER2	$\lambda=0$	0.214	0.256	0.320
17 venues, SPECTER2	$+w_{\text{cite}}=0.1$	0.317	0.324	0.396
29 venues, SPECTER2	$+w_{\text{cite}}=0.1$	0.324	0.327	0.388
17 venues, SPECTER2	$w_{\text{cite}}=0.1, w_{\text{vol}}=0.5$	0.263	0.303	0.350
17 venues, SPECTER2	$w_{\text{cite}}=0.1, w_{\text{sen}}=0.01$	0.292	0.309	0.351
17 venues, SPECTER2	$w_{\text{cite}}=0.1, h=8$	0.070	0.087	0.137

Table 2: Leave-one-out results, $n=200$, seed 0. The citation bonus gives the largest gain. The volume, seniority and recency terms lower this metric by design.

Stelmakh et al. (*A Gold Standard Dataset for the Reviewer Assignment Problem*) compared matching algorithms against self-reported reviewer expertise. SPECTER-family models on titles and abstracts with max pooling performed best. Full text gave TPMS a small gain. ORMatch belongs to that family. Full-text indexing is also limited here: 18% of the corpus has a direct open-access PDF, and INFORMS journals offer no text-mining access.

6 Limitations

- 37% of papers have no abstract. Almost all are Elsevier papers. An Elsevier API key would fill most of them.
- OpenAlex author disambiguation sometimes splits one person into several IDs or merges several people into one.
- PhD students appear only if they have a paper in an indexed venue.
- Many university and publisher sites block scripted access. Roster and board data are incomplete.
- All weights are set by hand. There are no elicited reviewer ratings to learn them from. The interface shows the objective with the current weights and lets the user change them.